



Team Details

- a. Team name: Code_With_Anik
- b. Team leader name: Anik Kumar Biswas- (Single contributor)
- c. Problem Statement: Prospect Assist AI – AI-Powered Retail Lending Decision Intelligence & Maker-Checker Platform
- d. Project Name: Project NIRNAY

Brief about the idea:

Project NIRNAY is an AI-powered Retail Lending Decision Intelligence Platform designed to help banks identify high-quality loan prospects, accurately estimate repayment capacity, and improve lending decisions through a transparent Maker–Checker approval workflow.

The solution combines machine learning, underwriting analytics, and explainable AI to analyze customer transaction behavior, credit profile, salary consistency, and repayment capability. It generates prospect scores, recommends suitable loan products, validates lending policies, and provides AI-assisted decision support while ensuring every approval passes through a secure Maker–Checker process with a complete audit trail.

By reducing manual effort, improving lead quality, and enabling transparent, policy-driven credit decisions, Project NIRNAY helps banks increase loan conversion rates while maintaining strong governance and regulatory compliance.

Opportunities

- ***How different is it from any of the other existing ideas?*** – Moves beyond traditional rule-based lending by combining AI-driven prospect scoring, income estimation, and behavioral transaction analysis. Integrates a Maker–Checker approval workflow, ensuring every lending decision is transparent, auditable, and compliant with banking governance.
- ***How will it be able to solve the problem?*** - Identifies high-quality loan prospects using customer transaction patterns, credit score, salary consistency, and repayment behavior. Estimates actual repayment capacity through underwriting metrics such as FOIR (Fixed Obligation to Income Ratio) and repayment capacity analysis. Recommends the most suitable loan product while validating applications against predefined lending policies.
- ***USP of the proposed solution*** - End-to-End AI-Powered Lending Decision Platform combining Prospect Scoring, Underwriting Intelligence, Explainable AI, Policy Validation, and Maker–Checker Governance. Transparent and auditable decision-making with a complete Audit Trail for regulatory compliance. Modular, scalable architecture that can be integrated with existing Core Banking Systems, Loan Origination Systems (LOS), and Credit Bureau services.

List of Features Offered by the Solution:

- **AI-Powered Prospect Scoring** – Identifies high-quality loan prospects using transaction behaviour, credit profile, and customer intent.
- 💰 **Intelligent Underwriting** – Estimates income, calculates FOIR and repayment capacity for prudent lending decisions.
- 🏠 **Smart Loan Recommendation** – Recommends the most suitable loan product (Personal, Home, Auto, Mortgage) based on customer eligibility.
- 🔍 **Explainable AI (XAI)** – Provides transparent reasons behind every AI recommendation to improve trust and decision-making.
- 📄 **Lending Policy Validation** – Automatically checks applications against predefined banking policies and eligibility criteria.
- ⚖️ **Risk & Eligibility Assessment** – Generates eligibility scores and risk classifications to support credit officers.
- 👥 **Maker–Checker Workflow** – Enables dual authorization with Maker and Checker approvals to ensure governance and compliance.
- 📁 **Audit Trail & Dashboard** – Maintains complete decision history and provides real-time insights through an interactive lending dashboard.

Process flow diagram or Use-case diagram :

Customer Data & Transactions



Data Preprocessing & Feature Engineering



AI Prospect Scoring & Income Estimation



Underwriting Assessment (*FOIR, Repayment Capacity, Eligibility Score*)



Lending Policy Validation & Risk Assessment



Explainable AI Recommendation (*Approve / Review / Reject*)

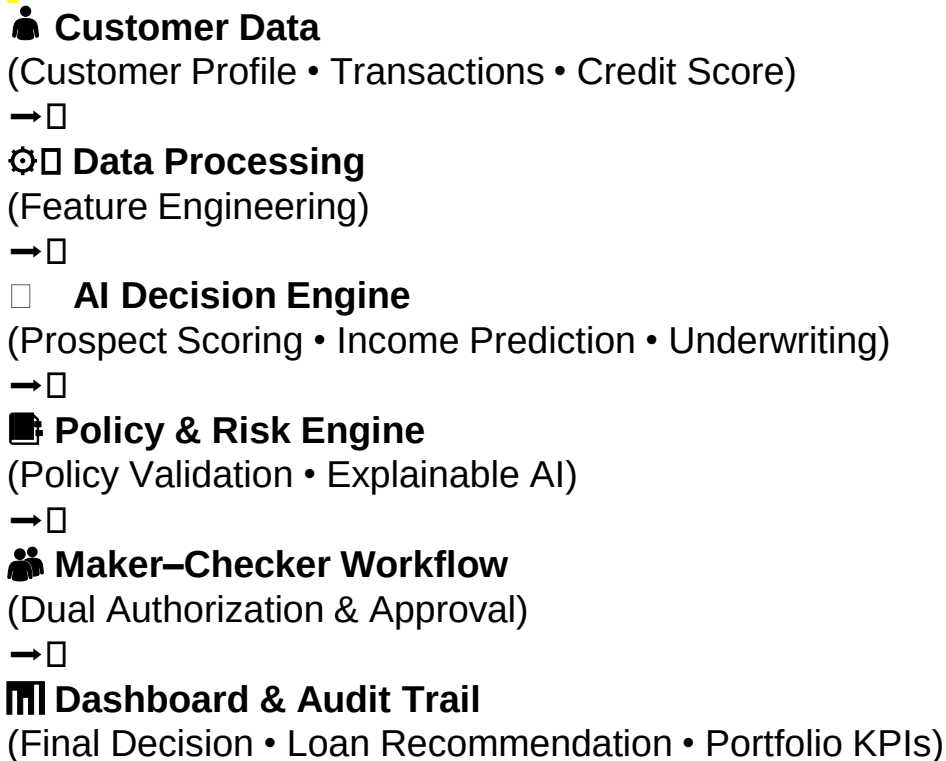


Maker–Checker Approval Workflow



Final Loan Decision, Audit Trail & Dashboard

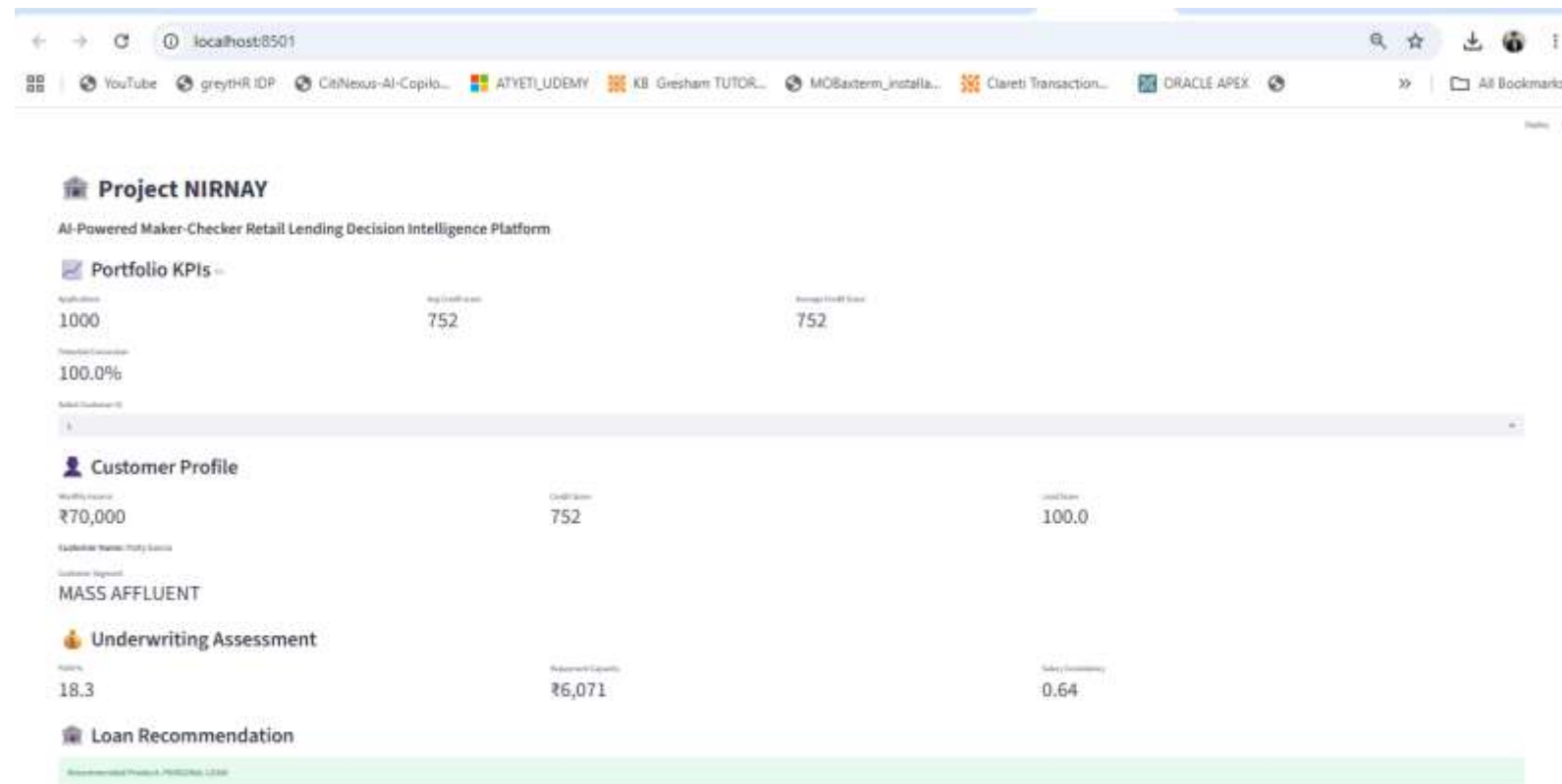
Architecture diagram of the proposed solution:



Technologies to be used in the solution:

Category	Technology
Programming Language	Python
Machine Learning	Scikit-learn, Pandas, NumPy
Data Generation	Faker
Frontend / Dashboard	Streamlit
Model Persistence	Joblib
Data Storage	CSV Files (Prototype)
Development Environment	PyCharm, Git, GitHub
Deployment	Streamlit Community Cloud / GitHub

Snapshots of the prototype:



Project NIRNAY
AI-Powered Maker-Checker Retail Lending Decision Intelligence Platform

Portfolio KPIs

Applications	Avg Credit Score	Average Credit Score
1000	752	752

Provision Conversion: 100.0%

Customer Profile

Monthly Income	Credit Score	Loan Score
₹70,000	752	100.0

Customer Segment: MASS AFFLUENT

Underwriting Assessment

Age	Payment Capacity	Debt to Income Ratio
18.3	₹6,071	0.64

Loan Recommendation

Recommended Product: PMSB2RBL_12.500

Prototype Performance report/Benchmarking:

Metric	Prototype Result
Prospect Scoring	Successfully identifies high-potential loan prospects
Income Prediction Model	R ² Score: 1.00 (<i>Synthetic Dataset</i>)
Income Prediction Error	MAE: 0.00 (<i>Synthetic Dataset</i>)
Underwriting Assessment	Automated FOIR & Repayment Capacity Calculation
Policy Validation	Real-time eligibility checks against lending rules
Decision Transparency	Explainable AI with decision reasons
Governance	Dual Maker–Checker approval with Audit Trail
User Interface	Interactive Streamlit Dashboard with real-time recommendations

Additional Details / Future Development :

Integrate with **Core Banking Systems (CBS)** and **Loan Origination Systems (LOS)** for end-to-end loan processing.

Connect to **Credit Bureau APIs** (e.g., CIBIL, Experian, CRIF) for real-time credit assessment.

Support **Bank Statement Analysis** using OCR and AI to improve income verification.

Implement **Fraud Detection & AML** models to identify suspicious lending activities.

Introduce a **GenAI Lending Assistant** to help credit officers summarize applications and explain decisions.

Add **Role-Based Access Control (RBAC)**, multi-level approvals, and digital signatures for enterprise deployment.

Deploy on a **secure cloud platform** with monitoring, logging, and scalable microservices architecture.

Enhance analytics with **portfolio monitoring, loan default prediction, and customer retention insights**.

Provide links to your:

- GitHub Public Repository : <https://github.com/Anik-1997-Biswas/IDBI-AI-Powered-Maker-Checker-Retail-Lending-Decision-Intelligence-Platform>
- Final Product Link:



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Thank you!